

CURRICULUM VITAE

EDUARDO MEDINA-ROLDAN

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Personal web page:

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An ecologist and educator with 20+ years of experience in ecology, and education. I have worked and adapted to different organisational cultures

“Believe those who are seeking the truth. Doubt those who say they’ve found it” (André Gide)

Interests: Soil-based ecosystem services and their synergies and trade-offs, microbial-driven soil processes, plant functional traits-soil links, land use change and intensification effects on soil resilience, global change ecology of arid and semiarid ecosystems, ecological theory of communities, ecological restoration, herbivore-plant-soil feedbacks, urban ecology, ecological indicators theory

PROFESSIONAL EXPERIENCE

- ❖ Research Fellow. National Research Council (CNR-IBE), Italy. EJP-Soil project: SERENA -Modelling Soil Functions and Soil Threats for Mapping Soil Functions and Ecosystem Services. **Description:** *pan-EU project -100 members- designing of methods to estimate soil ecosystem services and threats across different spatial scales via indicators and create user-friendly cookbooks for stakeholders.* 11/2021–05/2025.
- ❖ Lecturer. Department of Health and Environmental Sciences, Xi'an Jiaotong-Liverpool University (XJTLU-HES), China. **Description:** *All aspects of curriculum design and delivery at a high level of excellence, mentoring and supervising, classes from 7 to 100 students. Overseeing and leadership on the proper functioning of an academic department aligned with UK standards. Research on urban ecology and global change drivers effects on arid and semiarid ecosystems in collaboration with colleagues, mainly from China, outreach.* 08/2012–01/2021.
- ❖ Research Assistant. Run experiment on soil microbes and plant genetic diversity. San Luis Potosí Institute of Scientific Research and Technology (IPICYT). 02/2006–06/2006.
- ❖ Lab and field assistant. Project genetic diversity in grass populations in North American prairies. IPICYT. 01/2003–06/2003.

EDUCATION

- ❖ PhD. Biology. Lancaster University, UK. Thesis: *Herbivore-Plant Soil feedbacks: Their Role on Plant Resource Partitioning and the Organization and Functioning of a Grazing Ecosystem.* 11/2006–01/2012.
- ❖ MSc. Environmental Sciences and Natural Resource Management. IPICYT. Thesis: *Grazing effects on fungal root symbionts and C and N storage in a short-grass prairie in Central Mexico.* 08/2003-12/2005.
- ❖ BSc. Biological Sciences. University of Guadalajara, Mexico. Thesis: *Soil moisture dynamics across a range condition gradient in a short-grass steppe in Central Mexico (in Spanish).* 1997–2002.

PUBLICATIONS

Peer Reviewed International

- ❖ Qu, H., **E. Medina-Roldán**, X. Wang, M. Wang, S. Wang, X. Tang. Sustainability of domestic grazing is dependent on the vegetation state of arid grasslands: Evidence from 10 years of experimental grazing intensity observations. (*In review*).
- ❖ **Medina-Roldán, E.**, H. Huijser. Assessments to develop scientific enquiry and reflective practice in a Sino-British university (*In review*).
- ❖ **Medina-Roldán, E.**, L. Kukk, R. Lorenzetti, S. Pindral, F. Ungaro, Nicolas P.A. Saby, S. Asins, A. Astover, G. Bondi, G. Buttafuoco, J. Coblinski, E. Doñate, C. Foldal, R. Hessel, D. Montagne, J. Reyes-Rojas, L. O'Sullivan, I. Cousin. Empirical estimates of net ecosystem productivity for European wheat croplands based on spatially-explicit environmental data, and their use as a climate change regulation indicator (*Submitted*).
- ❖ Coblinski, J.A., S. Cornu, S. Pindral, J. Reyes-Rojas, **E. Medina-Roldán**, Lubos Borůvka, N.P.A. Saby. What threats for European soils by 2050? A bundle approach (*In review*).
- ❖ Qu, H., M. Wang, J. Wu; G. Wang; M. Zhang, G. Han; **E. Medina-Roldán**. Accelerating litter mass decomposition through the addition of mixed fungi in desert steppe (*In review*).
- ❖ Qu, H., M. Wang, L. Liu, S. Wang, K. Akhmedi, X. Ma, S. Zare, **E. Medina-Roldán**. Litter Addition Combined with Sand Barrier is an Effective Way on Degraded Desert Steppe Restoration. *Land Degradation and Development (Early View)*. [Click on doi](#).
- ❖ **Medina-Roldán, E.**, G. Buttafuoco, L. Gardin, R. Lorenzetti, F. Ungaro. 2026. Comparison of national and regional assessments of soil loss rates by water erosion and soil erosion control: an application to the Tuscany region (Italy). *Land* 15: 417. [Click on doi](#)

- ❖ Reyes-Rojas, J., D. Montagne, N.P.A. Saby, J.A. Coblinski, S. Pindral, **E. Medina-Roldán**, R. Lorenzetti, O. Scammacca, C. Piccini, L. Borůvka, and S. Cornu. 2025. Review of Soil Threats and Soil-Related Ecosystem Services European Maps: Can We Use Them to Study Their Relationships? *European J of Soil Science* 76:e70215. [Click on doi](#)
- ❖ Scammacca, O., D. Montagne, S. Asins, G. Bondi, L. Borůvka, G. Buttafuoco, A. Cadero, C. Calzolari, I. Cousin M. Czuba, C. Foldal, A. Malli, A. Klimkowicz-Pawlas, L. Kukk, E. Lumini, **E. Medina-Roldán**, K. Michel, M.J. Molina, L. O'Sullivan, S. Pindral, E. Putku, B. Kitzler, C. Walter. 2025. Assessing and mapping soil ecosystem services and soil threats changes in agroecosystems through scenario-based approaches – a systematic review. *Sci. Tot. Env.* 966:178646. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, M. Wang, T. Miyasaka, Y. Pan, X. Li, B. Liu, and H. Qu. 2024. Precipitation Controls Topsoil Nutrient Buildup in Arid and Semiarid Ecosystems. *Agriculture* 14:2364. [Click on doi](#)
- ❖ Song, Z., X. Zuo, Y. Hu, H. Wang, X. Zhang, Z. Wang, J. Qiao, L. Liu, and **E. Medina-Roldán**. 2024. Plant functional traits predict soil multifunctionality under increased precipitation and nitrogen addition in a desert steppe. *Catena* 247:108486. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, R. Lorenzetti, C. Calzolari, and F. Ungaro. 2024. Disentangling soil-based ecosystem services synergies, trade-offs, multifunctionality, and bundles: A case study at regional scale (NE Italy) to support environmental planning. *Geoderma* 448:116962. [Click on doi](#)
- ❖ Song, Z, X. Zhao, J. Qiao, Y. Hu, X. Li, P. Yue, M. Chen, S. Wang, **E. Medina-Roldán**. 2024. Plant functional traits mediate the response magnitude of plant-litter-soil microbial C: N: P stoichiometry to nitrogen addition in a desert steppe. *Sci. Tot. Env.* 915:169915. [Click on doi](#)
- ❖ Qu, H., **E. Medina-Roldán**, X. Zhao, S. Wang, X. Ma, X. Tang, L. Liu., 2024. A 5 and a half years long-term rainfall experiment shows clear precipitation thresholds in litter decomposition and litter nutrient dynamics in arid and semi-arid regions. *Biol. Fert. Soils* 1-14. [Click on doi](#)
- ❖ Sun, S., X. Liu, **E. Medina-Roldán**, H. Yuhui, P. Lv, Y. Xu. 2023. Precipitation regime changes alter allometry and plasticity of meristem allocation in annual herbaceous plants in a semiarid sandy grassland, Northern China. *Ecol. Res.* 38:628-638. [Click on doi](#)
- ❖ Guo, X., X. Zuo, **E. Medina-Roldán**, A. Guo, P. Yue, P., X. Zhao, J. Qiao, X. Li, M. Chen, C. Wei, and T. Yang. 2023. Effects of multi-resource addition on grassland plant productivity and biodiversity along a resource gradient. *Sci. Total Env.* 87:159367. [Click on doi](#)
- ❖ Li, X., X. Zuo, X. Zhao, S. Wang, P. Yue, C. Xu, Q. Yu, and **E. Medina-Roldán**, 2022. Extreme drought does not alter the stability of aboveground net primary productivity but decreases the stability of belowground net primary productivity in a desert steppe of northern China. *Env. Sci. & Pollution Res.* 30:4319-24328. [Click on doi](#)
- ❖ Sun, S., X. Liu, X. Zhao, **E. Medina-Roldán**, Y. He, P. Lv, and H. Hu. 2022. Annual herbaceous plants exhibit altered morphological traits in response to altered precipitation and drought patterns in semiarid Sandy grassland, Northern China. *Frontiers in Plant Sci.* 13:756950. [Click on doi](#)
- ❖ Qu, H., X. Zhao, X. Zuo, S. Wang, X. Ma, X. Tang, X. Wang, and **E. Medina-Roldán**. 2022. Litter decomposition in fragile ecosystems: A review. *Sciences in Cold & Arid Regions*, 14:151-161. [Click on doi](#)
- ❖ Wang, S., X. Zuo, T. Awada, **E. Medina-Roldán**, K. Feng, P. Yue, J. Lian, S. Zhao, and H. Cheng. 2021. Changes of soil bacterial and fungal community structure along a natural aridity gradient in desert grassland ecosystems, Inner Mongolia. *Catena* 205:105470. [Click on doi](#)
- ❖ Zuo X, W. Mao, H. Qu, M. Chen, S. Zhao, L. Liu, S. Wang, P. Yue, X. Ma, X. Zhao, **E. Medina-Roldán**. 2021 . Scale effects on spatial heterogeneity of herbaceous vegetation in desert steppe depend on plant community type. *Ecol. Ind.* 127:107769. [Click on doi](#)
- ❖ Lv, P., S. Sun, **E. Medina-Roldán**, S. Zhao, Y. Hu, A. Guo, X. Zuo. 2021. Effects of habitat types on the dynamic changes in allocation in carbon and nitrogen storage of vegetation–soil system in sandy grasslands: How habitat types affect C and N allocation?. *Ecol. & Evol.* 11:9079-9091. [Click on doi](#)
- ❖ Lv, P., S. Sun, **E. Medina-Roldán**, S. Zhao, Y. Hu, A. Guo, X. Zuo. 2021. Soil net nitrogen transformation rates are co-determined by multiple factors during the landscape evolution in Horqin Sandy Land. 2021. *Catena*. 206:105576.
- ❖ Zhao, Sh., T. Zhang, P. Yue, P. Lv, Y. Hu, **E. Medina-Roldán**, X. Zuo. 2021. Increased grazing intensities induce differentiation of the relationships between leaf traits and aboveground plant biomass in shrub- and grass-dominated community in desert steppe. *Ecol. Res.* 36:590-602. [Click on doi](#)
- ❖ Qu, Hao, X. Zhao, S. Wang; J. Lian, X. Tang, X. Wang, R. Zhang, and **E. Medina-Roldán**. 2021. Abiotic factors affect leaf litter mass loss more strongly than initial litter traits under sand burial conditions. *Catena* 196:104900. [Click on doi](#)
- ❖ Qu, Hao, X. Zhao, J. Lian, X. Tang, X. Wang, **E. Medina-Roldán**. 2020. Increasing precipitation interval has more impacts on litter mass loss than decreasing precipitation amount in desert steppe. *Frontiers Env. Sci.* 8:88. [Click on doi](#)
- ❖ Zhang, J., X. Zuo, X. Zhao, J. Ma, **E. Medina-Roldán**. 2020. Effects of rainfall manipulation and nitrogen addition on plant biomass allocation in a semiarid sandy grassland. *Sci. Rep.* 10:9026. [Click on doi](#)
- ❖ Hu, Y., X. Zuo, P. Yue, Sh. Zhao; X. Guo; X. Li, and **E. Medina-Roldán**. 2020. Increased precipitation shapes relationship between physiological and functional traits of *Stipa glareosa* in the grass- rather than shrub-dominated community in a desert steppe. *Plants* 9:1463. [Click on doi](#)
- ❖ Zuo, X., H. Chen, S. Zhao, Y. Ping, X. Liu, S. Wang, L. Liu, Ch. Xu, W-T. Luo, J.M.H. Knops, and **E. Medina-Roldán**. 2020. Observational and experimental evidence for the effect of altered precipitation on desert and steppe communities. *Global Ecol. & Conserv.* 21:e00864. [Click on doi](#)
- ❖ Rajasekar, A., R. Sekar, **E. Medina-Roldán**, J. Bridge, C. K. Moy, and S. Wilkinson. 2018. Biomineralisation performance of bacteria isolated from a landfill in China. *Canadian J. Microbiol.* 64:945-953.
- ❖ Rajasekar, A., R. Sekar, **E. Medina-Roldán**, J. Bridge, C.K. Moy, and S. Wilkinson. 2018. Next-generation sequencing showing potential leachate influence on bacterial communities around a landfill in China. *Canadian J, Microbiol.* 64:537-549. [Click on doi](#)

- ❖ Paz-González, A., C.A. de Abreu, A.M. Tarquis, and **E. Medina-Roldán**. 2014. Impacts of land use changes on soil properties and processes. Editorial. The Sci. World J. 2014:831975. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, E. Huber-Sannwald, and J.T. Arredondo. 2013. Plant phenotypic functional composition effects on soil processes in a semiarid grassland. Soil Biol. & Biochem. 66:1-9. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, and R.D. Bardgett. 2012. Inter-specific competition, but not differences in soil microbial communities, affects uptake of different chemical forms of nitrogen by graminoids of semi-natural grassland. PLoS ONE 7: e51193. [Click on doi](#)
- ❖ Paz-Ferreiro, J., **E. Medina-Roldán**, N.J. Ostle, N.P. McNamara, and R.D. Bardgett. 2012. Grazing increases the temperature sensitivity of soil respiration. Env. Res. Letters 7:014027. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, J. Paz-Ferreiro and R.D. Bardgett. 2012. Grazing-induced effects on soil properties modify plant competitive interactions in semi-natural mountain grasslands. Oecologia 170: 159-169. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, J. Paz-Ferreiro, and R.D. Bardgett. 2012. Grazing exclusion affects soil and plant communities, but has no impact on soil carbon storage in an upland grassland. Agr. Ecosyst & Env. 149:118-123. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, R.D. Bardgett. 2011. Plant and soil responses to defoliation: a comparative study of grass species with contrasting life history strategies. Plant & Soil 344:377-388. [Click on doi](#)
- ❖ **Medina-Roldán, E.**, T. Arredondo, E. Huber-Sannwald, L. Chapa Vargas, V. Olalde-Portugal. 2008. Grazing effects on fungal root symbionts and carbon and nitrogen storage in a shortgrass steppe in Central Mexico. J. Arid Env. 72:546-556. [Click on doi](#)
- ❖ Pineda-Martínez, L.F., J.N. Carbajal, and **E. Medina-Roldán**. 2007. Regionalization and classification of climate applying principal components analysis (PCA) at northeast region, México. Atmósfera 20:133-145. [Click on link](#)
- ❖ **Medina-Roldán, E.**, J.T. Arredondo, E. García-Moya, F.M. Huerta-Martínez. 2007. Soil water content dynamics along a range condition gradient in a shortgrass-steppe. Range Ecol. & Manage. 60:79–87. [Click on doi](#)

Conference Papers

- ❖ **Medina-Roldán, E.**, J. Paz-Ferreiro, J.T. Arredondo, E. Huber-Sannwald, R.D. Bardgett. Contrasting impacts of grazing on soil properties and plant communities between semiarid and temperate rangeland ecosystems. International Grassland Conference, Sydney, Australia. September 2013.

In preparation

- ❖ Coblinski, et al. Soil-based ecosystem services. A bundle approach.
- ❖ **Medina-Roldán, E.** Plant-soil feedbacks modify the microbial communities of bamboo plantations.
- ❖ **Medina-Roldán, E.** An ecological land use classification for rapidly changing urban areas.
- ❖ **Medina-Roldán, E.**, D. Hartono, and R. Sekar. Soil microbial communities' structure and diversity are weakly linked to urbanisation intensity in rapidly changing urban wetlands: A natural experiment approach.
- ❖ **Medina-Roldán, E.** Soil organic matter sensitivity linked to land use change across a land use and climatic gradient in central China.

CONFERENCES

- Coblinski, J., S. Cornu, N. Saby, J. Reyes-Rojas, **E. Medina-Roldán**, S. Pindral, R. Lorenzetti, C. Piccini, I. Cousin, L. Boruvka. Mapping of bundles of soil-based ecosystem services in the European Union and their evolution by 2050. Budapest Soil Health Forum, Dec. 2024, Budapest, Hungary.
- **Medina-Roldán, E.**, R. Lorenzetti (speaker), L. Kuk, S. Pindral, F. Ungaro, S. Asins, A. Astover, G. Bondi, G. Buttafuoco, J. Coblinski, E. Doñate, A. Fahy, C. Foldal, A. Matson Amanda, O. Mernagh, J. Reyes-Rojas, S. Motta, L. O'Sullivan, V. Shahrokh, E. Putku, E. Tagliaferri, E. Tonelli. An approach for mapping Net Ecosystem Productivity (NEP) as a pragmatic indicator of soil ecosystem service greenhouse gas (GHG) regulation including carbon sequestration in EU Member States. EJP-Soil Science Annual Days. June 2024. Vilnius, Lithuania (Lorenzetti speaker).
- Coblinski, J.A., S. Cornu, N. Saby, J. Reyes-Rojas, **E. Medina-Roldán**, S. Pindral, R. Lorenzetti, C. Piccini, L. Boruvka, I. Cousin. Bundles of soil-based ecosystem services in the European Union and their evolution by 2050. IUSS Centennial. May 2024. Florence, Italy.
- Lorenzetti, R., C. Calzolari, P. Tarocco, A. Aprea, **E. Medina-Roldán**, F. Ungaro. Indicators-based assessment of soil-based ecosystem services at different spatial scales in Emilia-Romagna (NE Italy). IUSS Centennial. May 2024. Florence, Italy (poster).
- Tarocco, P. A. Aprea, **E. Medina-Roldán (speaker)**, R. Lorenzetti, F. Ungaro. Challenges and opportunities to improve soil monitoring systems: the case of the Emilia-Romagna region in Italy. IUSS Centennial. May 2024. Florence, Italy.
- **Medina-Roldán, E.**, G. Buttafuoco, L. Gardin, R. Lorenzetti, and F. Ungaro. Comparison of national and regional assessments of soil loss rate by water erosion: an application to the Tuscany region (Italy). EGU 2024. Vienna, Austria.
- **Medina-Roldán, E.** How to Write an academic Paper? Henan University of Engineering, School of International Education Seminar. Sept. 2023. Zhengzhou, PR of China.
- Reyes-Rojas, J., J. Coblinski, N. Saby, I. Cousin, **E. Medina-Roldán**, L. Boruvka, S. Cornu. Mapping the spatial multifunctionality of soil-based ecosystem services relationships and bundles at European scale. Wageningen Soil Conference: Working together on Solutions for Sustainable World. Aug. 2023. Wageningen, the Netherlands.
- **Medina-Roldán, E.**, R. Lorenzetti, F. Ungaro, and C. Calzolari. Spatial relationships among soil-based ecosystem services can vary at the regional and local level. EGU. Apr. 2023. Vienna, Austria.
- Scammacca, O., A. Cadero, S. Asins, G. Bondi, L. Boruvka, G. Buttafuoco, C. Calzolari, M. Czuba, C. Foldal, A. Hofbauer, L. Kuk, E. Lumini, **E. Medina-Roldán**, K. Michel, M.J. Molina, L. O'Sullivan, S. Pindral, E. Putku, B. Kitzler, and C.

- Walter. What is the place for soils in scenario-based ecosystem services approaches? A systematic review. EGU. Apr. 2023. Vienna, Austria.
- Manna, P., A. Aprea, M. Bancheri, A. Basile, G. Buttafuoco, C. Calzolari, **E. Medina-Roldán**, P. Tarocco, and F. Ungaro. Multi-approach assessment of soil-based ecosystem services: the study case of Emilia Romagna Region (Italy). EGU. Apr. 2023. Vienna, Austria. (Poster).
 - **Medina-Roldán, E.**, R. Lorenzetti, P. Tarocco, F. Ungaro, C. Calzolari. Complementary approaches to capture spatial relationships among soil-based ecosystem services. Soil Mapping for a Sustainable Future. Feb. 2023. Orleans, France.
 - Asins, S., G. Bondi, L. Borůvka, G. Buttafuoco, A. Cadero, C. Calzolari, M. Czuba, C. Foldal, A. Hofbauer, B. Kitzler, L. Kukkk, E. Lumini, **E. Medina-Roldán**, K. Michel, M.J. Molina, L. O'Sullivan, S. Pindral, E. Putku, O. Scammacca, C. Walter. Using digital soil mapping data from scenario studies to model changes in soil threats and soil ecosystem services - a meta-analysis. Soil Mapping for a Sustainable Future. Feb. 2023. Orleans, France.
 - **Medina-Roldán, E.** (Session Organiser). Plant soil Interactions and Sustainability. Session for Mexican Congress of Soil Science. Oct. 2022. Saltillo, Mexico
 - **Medina-Roldán, E.** Effects of herbivores on the competition of plants through plant-soil feedbacks (In Spanish). Mexican Congress of Soil Science. Saltillo, Mexico, Oct. 2022.
 - Calzolari C., R. Lorenzetti, **E. Medina-Roldán**, P. Tarocco, F. Ungaro. Soil ecosystem service bundles at regional scale: A Case Study from Emilia-Romagna (NE Italy). EJP-Soil Annual Days. June 2022. Palermo, Italy.
 - Lorenzetti, R., **E. Medina-Roldán**, L. Gardin, G. Buttafuoco, M. Fantappiè, F. Assennato, E. Tondini, and S. Milena. The data behind soil threats and soil ecosystem services: Italian state of the art. EJP-Soil Annual Science Days. June 2022. Palermo, Italy. (Poster).
 - **Medina-Roldán, E.** Herbivore-plant-soil interactions in semiarid and upland grasslands and other stories. NIOO-Knaw Seminar. Feb. 2020. Wageningen, the Netherlands.
 - **Medina-Roldán, E.** Herbivore-plant-soil interactions in semiarid and upland grasslands and other stories. Northwest Institute of Eco-environment and Resources, CAS. Seminar. Nov. 2019. Lanzhou, PR of China.
 - **Medina-Roldán, E.**, D. Hartono, and R. Sekar. Urbanisation Intensity Increases Soil Microbial Richness, But It Is Not Linked To Microbial Community Structure: A Natural Experiment In Rapidly Changing Urban Wetlands. SUITMA 10th Conference. June 2019. Seoul, South Korea.
 - **Medina-Roldán, E.** The patchwork assessment: a formative-summative assessment combo in an ecology module. XJTU Annual Learning and Teaching Colloquium - Innovative Teaching and Student Learning. Apr. 2017. Suzhou, PR of China.
 - **Medina-Roldán, E.**, J. Grisolia, M.E. Macken, and Y. Zhang. Making the Concept of Ecosystem Services "Real": An Interdisciplinary View from Environmental Sciences, Visual Arts and Economy. Society for Urban Ecology 2nd International Congress. July 2016. Shanghai, PR of China.
 - **Medina-Roldán, E.**, Wu, Y. Urban parks as models to test ecological theories. 12th ESAFS Conference, Sept. 2015. Nanjing, PR of China.
 - **Medina-Roldán, E.**, Jorge Paz-Ferreiro, Richard D. Bardgett. Hidden effects of large herbivores on plant competition through plant-soil feedbacks. First Global Soil Biodiversity Conference. December 2014. Dijon, France.
 - **Medina-Roldán, E.**, T. Arredondo, E. Huber-Sannwald. Phenotypic composition of a key grass species and soil processes in a semiarid grassland after a rain pulse event. 20th World Congress of Soil Science. June 2014. Jeju, South Korea.
 - **Medina-Roldán, E.**, J. Paz-Ferreiro, J.T. Arredondo, E. Huber-Sannwald, R.D. Bardgett. Contrasting impacts of grazing on soil properties and plant communities between semiarid and temperate rangeland ecosystems. International Grassland Conference. Sept. 2013. Sydney, Australia. Conference paper.
 - **Medina-Roldán, E.** Recognizing multiple grazing impacts to use it as a management tool: the belowground perspective. SCOPE-ZHONGYU Environmental Forum. Oct. 2012. Taiyuan, PR of China.
 - **Medina-Roldán, E.** Plant and soil responses to defoliation: contrasting grasses with different life history strategies under nutrient limitation. Symposium of The UK's MexSoc. July 2010. Manchester, UK.
 - **Medina-Roldán, E.** Grazing effects on soil biogeochemistry in Northern England uplands. Symposium of The UK's MexSoc. Apr. 2009. Cambridge, UK.
 - **Medina-Roldán, E.** Tales of cowboys, rangelands and ecosystem functioning: the case of the shortgrass prairie. Lancaster Environment Centre Seminar Series. Feb. 2009. Lancaster, UK.
 - **Medina-Roldán, E.** Above-belowground interactions in N-limited upland ecosystems in Northern England. Lancaster Environment Centre Seminar Series. May 2008. Lancaster, UK.
 - Huber-Sannwald, E., T. Arredondo, and **E. Medina-Roldán**. Nitrogen mineralization in response to a water pulse: The role of phenotypic differences in plant carbon quality on belowground processes. 92nd ESA Annual Meeting. Dec. 2007. San Jose, California, US.
 - **Medina-Roldán, E.** and R.D. Bardgett. Soil and plant responses to defoliation among temperate grasses differing in relative growth rates. British Ecological Society Annual Meeting. Sept. 2009. Hertfordshire, UK.
 - **Medina-Roldán, E.**, J. Paz-Ferreiro, R.D. Bardgett, N.P. McNamara, N.J. Ostle. Ecological above-belowground interactions drive a positive feedback maintaining nutrient limitation in upland vegetation in Northern England. British Ecological Society Annual Meeting. Sept. 2008. London, UK.
 - Paz-Ferreiro, J., **E. Medina-Roldán**, R.D. Bardgett, N.J. Ostle, N.P. McNamara. Dependence of CO₂ emissions with fertilizer input, grazing and temperature in an upland grassland. British Ecological Society Annual Meeting. Sept. 2008. London, UK.
 - Arredondo, T., E. Huber-Sannwald, **E. Medina-Roldán**. Differential control on soil carbon and water content among genotypes. Global Environmental Change - Regional Challenges. An Earth System Science Partnership (ESSP). Nov. 2006. Beijing, China.
 - **Medina-Roldán, E.**, A. Jiménez-Aguilar, L.F. Pineda-Martínez, L. Hernández-Azcúnaga, J.T. Arredondo, and E. Huber-Sannwald. Carbon stocks under different land use change and land cover scenarios: A case study in the watershed "San Miguelito" San Luis Potosí, Mexico. Carbon Management at Urban and Regional Levels: Connecting Development Decisions to Global Issues. Sept. 2006. Mexico City, Mexico
 - **Medina-Roldán, E.**, T. Arredondo, E. Huber-Sannwald. Grazing effects on root fungal symbionts and soil biogeochemical properties in semiarid grasslands of Central Mexico. Ecology in an Era of Globalization, ESA Conference. Jan. 2006. Merida, Mexico.

- Arredondo, T., E. Huber-Sannwald, **E. Medina-Roldán**. Soil water dynamics of model populations of *Bouteloua gracilis* differing in lignin content. 90th Annual Meeting of the Ecological Society of America. Aug. 2005. Montreal, Canada.
- Arredondo, T., **E. Medina-Roldán**, E. Huber-Sannwald. Overgrazing as a factor of land use change in semiarid grasslands of Central Mexico. Land Open Science Conference. December, 2003. Morelia, Mexico.
- Huber-Sannwald, E., T. Arredondo, **E. Medina-Roldán**, A. Jiménez-Aguilar. A regional analysis of land degradation as a basis for the selection of restoration sites in Central Mexico. 15th Annual Conference of the Society for Ecological Restoration International. Nov. 2003. Austin, Texas, US.
- ARIDNET. Interdisciplinary workshop on Land degradation in semiarid regions of the Americas - The Amapola, Mexico Case Study, San Luis Potosí, Mexico. June 2004.
- Vitae, UK. The engaging researcher, Manchester, UK. December, 2010.

RESEARCH GRANTS/PROJECT COLLABORATIONS

- ❖ General Project of National Natural Science Foundation of China (42377473): Nutrient Release Effects and Driving Mechanisms of Plant Litter in Desert Grasslands in the Upper Yellow River
- ❖ General Project of National Natural Science Foundation of China (41877540): Comparative study on the response of litter decomposition to changes in precipitation patterns in ecologically fragile arid and semi-arid areas
- ❖ National Key Research and Development Program Project of China (2016YFC0500506): Research and demonstration of degraded grassland restoration and ecological animal husbandry technology in desert areas
- ❖ Key Research and Development Program Project of Gansu Province(25YFWA008): Collaborative research and application of organic waste resource utilization technology in agriculture and animal husbandry between China and Mongolia
- ❖ Key Research and Development Program Project of Ningxia Province (2020BBF02003): Research on Key Technologies and Model Creation for Enhancing the Function of Grass and Livestock Industry in Yanchi County Based on Water Resource Carrying Capacity
- ❖ Urbanisation and heavy metal pollution in a warmer world. Xi'an Jiaotong-Liverpool University Graduate Postgraduate Research Grant (PI, in collaboration with several colleagues). 200,000 RMB.
- ❖ Stimulated microbial activity for sustainable geotechnical remediation of unlined/uncontrolled municipal solid waste disposal sites. Xi'an Jiaotong-Liverpool University Research Development Fund (in collaboration with several colleagues). 100,000 RMB.
- ❖ Soil carbon fluxes and stability of soil biological properties in response to intensified drought: plant species and restoration of degraded arid areas in Chile. Approved project in the 2013 post-doctoral call of the FONDECYT (Scientific and Technological National Development Fund, Chile). Funds worth \$ 92,000 US.

RESEARCH STAYS

- ❖ Info&Sols Unit, INRAE, Orleans, France. Host: Nicola Saby, Isabelle Cousin, Antonio Bispo. 06/2024.
- ❖ NIOO-KNAW(Netherlands Institute of Ecology), Wageningen, Netherlands. Host: Prof. Wim van der Putten. 12/2019–02/2020.
- ❖ Northwest Institute of Eco-Environment and Resources, Chinese Academy of Sciences, Lanzhou, China. Host: Prof. Xiaolan Zuo. 10/2029–12/2019.
- ❖ INIFAP. Professional Practices (herbarium and drylands farming/livestock outreach). Jalisco, Mexico. 04/2002–09/2002.

HONOURS AND AWARDS

- ❖ Guest Expert, State Key Laboratory of Ecological Safety and Sustainable Development in Arid Lands, Urat Desert-grassland Research Station, Northwest Institute of Eco-Environment and Resources, Chinese Academy of Sciences, Lanzhou, 730000, China (2025–current).
- ❖ Grant for PhD studies. Council for Science and technology (CONACYT) (2006–2010).
- ❖ Grant for Master studies. CONACYT (2003–2005).
- ❖ Grant for Bachelor thesis development. IPICYT (2003–2003).
- ❖ Grant for Bachelor thesis development. CONACYT (2001–2002).

ADMINISTRATIVE EXPERIENCE

- ❖ SERENA project. EJP-Soil. Task coordination, whole-project admin support. 2021–2024.
- ❖ PhD Programme Director. Programme framework, liaison and students mentoring. XJTLU-HES. 2017–2021.
- ❖ Research Seminars Coordinator. XJTLU-HES. 2017–2019.
- ❖ Academic Practice. Quality standards in Learning & Teaching. XJTLU-HES. 2017.
- ❖ Research Officer. Coordination department's research needs and support. XJTLU-HES. 2016–2017.
- ❖ Exam Officer. Quality assurance on assessments based on UK system standards. XJTLU-HES. 2015–2016.
- ❖ Learning & Teaching Officer. Compliance quality in learning and teaching, annual programme reviews, Learning & Teaching policy. XJTLU-HES. 2012–2015.
- ❖ Student-Staff liaison Committees (Yr1, Yr2, PhD). Two-way communication channel students/staff. XJTLU-HES. 2015–2018.

TEACHING EXPERIENCE

Modules designed and taught at University level

Module (Yr)	Dates/Institution	Role	Highlights
Planet Earth (Y1)	09/2023, Zhejiang University of Engineering, China	Module leader (ML)	Designed & delivered for students with a basic level of English, smartphone quizzes to support learning
Ecosystem services and natural capital	03/2023, Universidad Autonoma Metropolitana, Mex.	Guest Lecturer	
Intro. Planet Earth (Y1)	2020, Xi'an Jiaotong-Liverpool University (XJTLU), China	ML	Designed & delivered. Hands-on practical collaborative assessments including one on climate system modelling with online tool
Soil and the Environment (Y2)	2013–2019, XJTLU	ML	Designed & delivered. Students self and peer assessment, co-creation of marking criteria, paper peer review, soil studio for in-class experiments, reciprocal teaching for reading primary sources; soil id field trip together with ISSAS, Nanjing
Ecology and Conservation (Y1)	2013–2020, XJTLU	ML	Designed & delivered. Patchwork text for scientific enquiry, reflective diary, role play on socio-scientific issues, peer- and self-assessments, on-line forums, video essays/video-feedback, games for ecological theory, cemetery field practice
Field Trip Skills Intro (Y1)	2013–2016, 2020, XJTLU	Co-delivered (CO)/ML	Terrestrial ecology field skills, 2020 Adjustment to e-learning with focus on collaborative home experiment for data acquisition, and plant id using online resources
Final Year Project (Y3)	2016–2017, XJTLU	ML/CO	Marking criteria for dissertations, presentations and proposals; skills development framework, supervision of FYPs
Aquatic and Urban Ecology (Y2)	2013–2017, XJTLU	CO	Designed & delivered Urban Ecology. Interdisciplinary urban ecosystem services assessment
GIS and Spatial Modelling (Y2)	2016–2017, XJTLU	CO	Designed & delivered spatial ecology. Modelling in R, practical use of GIS/RS in QGIS/GRASS. Collaborative environmental consultancy problem
Research Skills (Y2)	2013–2020, XJTLU	CO	Designed & delivered. Scientific method know-how and hands-on scientific writing
Research Skills and GIS (Y1)	2012–2013, XJTLU	CO	Designed & delivered. Hands-on practicals using GRASS/QGIS
Field Trip Skills Advanced (Y2)	2015, XJTLU	CO	Hands on practical on soil erosion/restoration, students' data collection
Environmental Chemistry (Y1)	2012, XJTLU	ML	Designed & delivered. Adjustment for students with low chemistry training
Dynamic Landscapes (Y1)	2012–2013, 2014-2015, XJTLU	CO	Co-Designed. Geo-morphological topics with in class studios

Global Environmental Issues (Yo)	2015–2018, XJTLU.	CO	Cross-university module to introduce relevant environmental topics, focus on biodiversity and evolution
Communication, Business and the Environment (Y1)	2016–2020, XJTLU	CO	Cross-university module to introduce environmental topics using a CLIL approach, focus ecosystem services and natural capital
The Anthropocene for journalism students (Y2)	2018, XJTLU	Guest Lecturer	
ATLAS week (Yo)	2016-2017, XJTLU	Coord	Adoption of sustainability activities for the intro week for freshers

Supervision and Mentoring

- ❖ PhD (co-supervisor). Stimulated microbial activity for sustainable geotechnical remediation of unlined/uncontrolled municipal solid waste disposal sites. Completed. XJTLU.
- ❖ UG Final Year Projects in XJTLU-HES for over 8 years.
- ❖ Academic adviser students BSc in Environmental Sciences. XJTLU-HES. 2012–2021.

TEACHING QUALIFICATIONS

- ❖ Higher Education Academy, UK fellow. 2016.
- ❖ Certificate in Professional Studies in Learning and Teaching in Higher Education. University of Liverpool/XJTLU. 2016.

Modules QC/QA as Internal Moderator (2012-2021): Assessments, structure, marking

UG Year 1	UG Year 2	UG Year 3
•Biodiversity of Animals, Plants & Microbes	•Environmental Change: Past, Present & Future	•Environmental Management
•Environmental Chemistry	•Research Design	•Environmental Impact Assessment
•Aquatic Field Skills		•Ecology in a changing world
•Dynamic Landscapes		
•Environmental Pollution		

Programmes QC/QA as Internal Moderator (2012-2021)

- ❖ New Programme in MSc. Operations and Supply Chain Management (International Business School, XJTLU)
- ❖ BSc. Environmental Sciences Program Review, Yr 2 coordinator (XJTLU-HES)
- ❖ New Programme in MDes Transdisciplinary Design (Department of Industrial Design, XJTLU)
- ❖ New Programme in BSc in Environmental Science, field-based modules designer (XJTLU-HES)

OTHERS

Peer Review Experience

- ❖ Title include: Nature Sustainability, Global Change Biology, Plant & Soil, Applied Soil Ecology, Biology & Fertility of Soils, Ecosystems, Canadian Journal of Botany, PLoS One, Ecology, Oecologia, Journal of

Ecology, Acta Oecologica, Functional Ecology, Ecological Applications, Ecological Monographs, Geoderma, Catena, Soil Biology & Biogeochemistry, among others

Editorial Work

- ❖ Guest Editor. Nutrient Cycling and Management in Drylands: Challenges and Innovations in Arid and Semi-Arid Regions. Lands. 2025.
- ❖ Guest Editor. Soil and Land Use Change. The Scientific World Journal. 2014.

Languages

- ❖ English, advanced writing, speaking, listening, and reading
- ❖ Italian (B2), writing, speaking, listening, and reading
- ❖ Spanish (mother tongue)

Other Qualifications

- ❖ Qualified Mountain Guide, Piemonte Region, Italy.
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